$\boxed{\text{inclusion-exclusion identity}}$

1. $P(E^c) = 1 - P(E)$

proof: $1 = P(S) = P(E \cup E^c) = P(E) + P(E^c)$

$\therefore P(E^c) = 1 - P(E)$

2. $E \subset F \Rightarrow P(E) \leq P(F)$

proof: $\because F = E \cup (F \setminus E) = E \cup (E^c F)$

$\therefore P(F) = P(E) + P(E^c F)$

$\therefore P(F) \geq P(E)$

3. $P(E \cup F) = P(E) + P(F) - P(EF)$

proof: $\because E \cup F = E \cup E^c F$

$\therefore P(E \cup F) = P(E) + P(E^c F)$

$\because F = EF \cup E^c F$

$\therefore P(F) = P(EF) + P(E^c F)$

$\therefore P(E^c F) = P(F) - P(EF)$

$\therefore P(E \cup F) = P(E) + P(F) - P(EF)$

4. $P(E_1 \cup E_2 \cup E_3)$

$= P(E_1) + P(E_2) + P(E_3)$

$- P(E_1E_2) - P(E_1E_3) - P(E_2E_3)$

$+ P(E_1E_2E_3)$

proof: $P(E_1 \cup E_2 \cup E_3)$

$= P((E_1 \cup E_2) \cup E_3)$

$= P(E_{12}) + P(E_3) - P(E_{12}E_3)$

here $E_{12} = E_1 \cup E_2$

$\because P(E_{12}) = P(E_1) + P(E_2) - P(E_1E_2)$

$P(E_{12}E_3) = P(E_1E_3 \cup E_2E_3)$

$= P(E_1E_3) + P(E_2E_3)$

$- P(E_1E_2E_3)$

$\therefore P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)$

$- P(E_1E_2) - P(E_1E_3) - P(E_2E_3)$

$+ P(E_1E_2E_3)$

$\blacksquare$

5. inclusion-exclusion identity

$$P(E_1 \cup E_2 \cup \cdots \cup E_n) = \sum_{i=1}^{n} P(E_i) - \sum_{i_1 < i_2} P(E_{i_1}E_{i_2})$$

$$+ \cdots + (-1)^{r+1} \sum_{i_1 < i_2 < \cdots < i_r} P(E_{i_1}E_{i_2} \cdots E_{i_r})$$

$$+ \cdots + (-1)^{n+1} P(E_1E_2 \cdots E_n)$$

i.e, $$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{r=1}^{n} (-1)^{r+1} \sum_{i_1 < i_2 < \cdots < i_r} P(E_{i_1}E_{i_2} \cdots E_{i_r})$$

5.3 proof: (method 1)

(mathematical induction)

① $n = 1, 2, 3$, left = right

② assume $l = r$ for $n-1$

③ $P\left(\bigcup^{n} E_i\right) = P\left(\left(\bigcup^{n-1} E_i\right) \cup E_n\right)$

$= P\left(\bigcup^{n-1} E_i\right) + P(E_n) - P\left(\left(\bigcup^{n-1} E_i\right) E_n\right)$

$\because P\left(\bigcup^{n-1} E_i\right) = \sum_{r=1}^{n-1} (-1)^{r+1} \sum_{i_1 < i_2 < \cdots < i_r} P(E_{i_1}E_{i_2} \cdots E_{i_r})$

$= (\bigcup^{n-1} E_i) \cup (E_1^c E_2^c \cdots E_{n-1}^c) E_n$

$= E_1 \cup E_1^c E_2 \cup \cdots \cup E_1^c E_{n-2}^c E_{n-1} \cup E_1^c E_2^c E_{n-1}^c E_n$

$\therefore l = r$ ∎

7. $P(\bigcup_{i=1}^{n} E_i) = \sum_{i=1}^{n} P(E_i) - \sum_{i=2}^{n} P(\bigcup_{j:j<i} E_i E_j)$, $n \geq 1$

proof: ① $\bigcup_{i=1}^{n} E = E_1 \cup E_1^c E_2 \cup \cdots \cup E_1^c E_{n-1}^c E_n$

$\because$ right is union of disjoint events

$\therefore P(\bigcup_{i=1}^{n} E) = P(E_1) + \sum_{i=2}^{n} P(E_1^c E_{i-1}^c E_i)$

② $\forall B$, $P(E_i) = P(BE_i) + P(B^c E_i)$

$\therefore P(E_i) = P(E_1^c E_{i-1}^c E_i) + P((E_1^c E_{i-1}^c)^c E_i)$

$= P(E_1^c E_{i-1}^c E_i) + P((\bigcup_{j=1}^{i-1} E_j) E_i)$

$= P(E_1^c E_{i-1}^c E_i) + P(\bigcup_{j:j<i} E_i E_j)$

$\therefore P(E_1^c E_{i-1}^c E_i) = P(E_i) - P(\bigcup_{j:j<i} E_i E_j)$

③ $\therefore P(\bigcup_{i=1}^{n} E_i) = P(E_1) + \sum_{i=2}^{n} P(E_i) - \sum_{i=2}^{n} P(\bigcup_{j:j<i} E_i E_j)$

$\therefore P(\bigcup_{i=1}^{n} E_i) = \sum_{i=1}^{n} P(E_i) - \sum_{i=2}^{n} P(\bigcup_{j:j<i} E_i E_j)$, $n \geq 1$ ∎.

5.4 method2, see 10.

6. $\bigcup_{i=1}^{n} E_i = E_1 \cup E_1^c E_2 \cup E_1^c E_2^c E_3 \cup \cdots \cup E_1^c E_2^c E_{n-1}^c E_n$,

proof: ① $n=1$, $l=r$.

$n=2$, $l = E_1 \cup E_2$

$= E_1 \cup (E_2 \setminus E_1)$

$= E_1 \cup E_1^c E_2 = r$

$n=3$, $l = E_1 \cup E_2 \cup E_3$

$= (E_1 \cup E_2) \cup (E_3 \setminus (E_1 \cup E_2))$

$= (E_1 \cup E_2) \cup (E_1 \cup E_2)^c E_3$

$= (E_1 \cup E_2) \cup E_1^c E_2^c E_3$

$= E_1 \cup E_1^c E_2 \cup E_1^c E_2^c E_3$

② if $l=r$ for $n-1$.

③ $\bigcup_{i=1}^{n} E_i = (\bigcup_{i=1}^{n-1} E_i) \cup E_n$

$= (\bigcup^{n-1} E_i) \cup (\bigcup^{n-1} E_i)^c E_n$

$P((\bigcup^{n-1} E_i) E_n) = P(\bigcup_{i=1}^{n-1} (E_i E_n))$

$= \sum_{r=1}^{n-1} (-1)^{r+1} \sum_{i_1 < i_2 < \cdots < i_r < n} P(E_{i_1} E_{i_2} \cdots E_{i_r} E_n)$

$\therefore P(E_n) - \sum_{r=1}^{n-1} (-1)^{r+1} \sum_{i_1 < i_2 < i_r < n} P(E_{i_1} E_{i_r} E_n)$

$= P(E_n) + \sum_{r'=2}^{n} (-1)^{r'+1} \sum_{i_1 < i_2 < i_{r'-1} < i_{r'} = n} P(E_{i_1} \cdots E_{i_{r'}})$

$= \sum_{r=1}^{n} (-1)^{r+1} \sum_{i_1 < i_{r-1} < i_r = n} P(E_{i_1} \cdots E_{i_r})$

$\therefore P(\bigcup_{i=1}^{n} E_i) = \sum_{r=1}^{n-1} (-1)^{r+1} \sum_{i_1 < i_r < n} P(E_{i_1} E_{i_2} E_{i_r})$

$+ \sum_{r=1}^{n} (-1)^{r+1} \sum_{i_1 < i_2 < \cdots < i_{r-1} < i_r = n} P(E_{i_1} \cdots E_{i_r})$

$= \sum_{r=1}^{n} (-1)^{r+1} \sum_{i_1 < \cdots < i_r} P(E_{i_1} E_{i_2} E_{i_r})$

∎

8.0

$$P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_i P(E_i) - \sum_{i=2}^{n} P\left(\bigcup_{j:j<i} E_iE_j\right)$$

$$= \sum_{i=1}^{n} P(E_i) - \sum_{i=2}^{n} P\left(\bigcup_{j=1}^{i-1} E_iE_j\right)$$

$$= \sum_{i=1}^{n} P(E_i) - \sum_{i=1}^{n} P\left(\bigcup_{j=1}^{i-1} E_iE_j\right)$$

(here $\bigcup_{j=1}^{i-1} E_iE_j = \emptyset$ if $i=1$)

$$= \sum_{i=1}^{n} P(E_i) - \sum_{i=1}^{n} P\left(\bigcup_{j:j<i} E_iE_j\right), \ n\in Z$$

(if $n\le 0$, $r=l=0$)

8.1 $P\left(\bigcup_{i=1}^{n} E_i\right) \le \sum_{i=1}^{n} P(E_i)$, $n\in Z$

proof: ① ∵ $\sum_{i=1}^{n} P\left(\bigcup_{j:j<i} E_iE_j\right) \ge 0$

② from 8.0, $P\left(\bigcup E_i\right) = \sum P(E_i) - \sum_{i=1}^{n} P\left(\bigcup_{j:j<i} E_iE_j\right)$

∴ $P\left(\bigcup E_i\right) \le \sum P(E_i)$, $n\ge 1$

③ if $n\le 0$, $\bigcup_{i=1}^{n} E_i := \emptyset$

$\sum_{i=1}^{n} P(E_i) := 0$

∴ $P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} P(E_i) = 0$

8.2 $P\left(\bigcup_{i=1}^{n} E_i\right) \ge \sum_{i=1}^{n} P(E_i) - \sum_{i,j:j<i} P(E_iE_j)$

proof: ① use 8.1,

$P\left(\bigcup_{j:j<i} E_iE_j\right) \le \sum_{j:j<i} P(E_iE_j), \forall i\in Z$

② from 8.0,

$$P\left(\bigcup E_i\right) = \sum P(E_i) - \sum_{i=1}^{n} P\left(\bigcup_{j:j<i} E_iE_j\right)$$

$$\ge \sum P(E_i) - \sum_{i=1}^{n} \sum_{j:j<i} P(E_iE_j)$$

$$= \sum P(E_i) - \sum_{i,j:j<i} P(E_iE_j)$$

(note $\sum_{i,j:j<i} P(E_iE_j) = 0$, if $n<2$)

8.3 $P\left(\bigcup_{i=1}^{n} E_i\right)$

$$\le \sum_{i=1}^{n} P(E_i) - \sum_{i,j:j<i} P(E_iE_j) + \sum_{i,j,k:k<j<i} P(E_iE_jE_k), \ n\in Z$$

proof: ① use 8.2,

$\forall i\in Z$, $P\left(\bigcup_{j:j<i} E_iE_j\right) \ge \sum_{j:j<i} P(E_iE_j) - \sum_{k,j:k<j<i} P(E_iE_kE_j)$

② from 8.0, $P\left(\bigcup_{i=1}^{n} E_i\right) = \sum_i P(E_i) - \sum_{i=1}^{n} P\left(\bigcup_{j:j<i} E_iE_j\right)$

$$\le \sum_i P(E_i) - \sum_{i=1}^{n}\left(\sum_{j:j<i} P(E_iE_j) - \sum_{k,j:k<j<i} P(E_iE_jE_k)\right)$$

$$= \sum_i P(E_i) - \sum_{i,j:j<i} P(E_iE_j) + \sum_{i,j,k:k<j<i} P(E_iE_jE_k), \ n\in Z$$

(note $\sum_{i,j,k:k<j<i} P(E_iE_jE_k) = 0$ if $n<3$)

$$= \sum_{i_1=1}^{n} P(E_{i_1}) - \sum_{i_1=1}^{n}\sum_{r'=2}^{j+1}(-1)^{r'} \sum_{i_2,i_3,i_{r'}:\, i_{r'}<i_3<i_2<i_1} P(E_{i_1}E_{i_2}E_{i_{r'}})$$

$$= \sum_{i_1=1}^{n} P(E_{i_1}) + \sum_{r'=2}^{j+1}(-1)^{r'+1} \sum_{i_1,i_2,i_{r'}:\, i_{r'}<i_2<i_1} P(E_{i_1}E_{i_2}E_{i_{r'}})$$

$$= \sum_{r=1}^{j+1}(-1)^{r+1} \sum_{i_1,i_2,i_r:\, i_r<i_2<i_1} P(E_{i_1}E_{i_2}E_{i_r})$$

$\therefore$ for $k=j+1$, 9.1 hold.

$\therefore$ $\forall k\in Z^+$, 9.1 hold. ▨

10. $P(\bigcup E_i) = \sum_{r=1}^{n}(-1)^{r+1} \sum_{i_1,i_2,i_r:\, i_r<i_2<i_1} P(E_{i_1}E_{i_2}E_{i_r})$

proof: (method 2 of 5)

for 9.1, $\because k\in Z^+$

$\therefore$ let $k=n \Rightarrow l \ (-\leq)^{n+1}\ r$

let $k=n+1 \Rightarrow l \ (-\leq)^{n+2}\ r$

$\Rightarrow l=r$

▨

if: $k=j \Rightarrow P(\bigcup_{i=1}^{n} E_i)\ (-\leq)^{j+1} \sum_{r=1}^{j}(-1)^{r+1} \sum_{i_1,i_2,i_r:\, i_r<i_2<i_1} P(E_{i_1}E_{i_2}E_{i_r})$

$\forall n\in Z$

then: $\forall i_1 \in Z$,

$$P\Big(\bigcup_{i_2:\, i_2<i_1} E_{i_1}E_{i_2}\Big) = P\Big(\bigcup_{i_2=1}^{i_1-1} E_{i_1}E_{i_2}\Big)$$

$$= P\Big(\bigcup_{h=1}^{i_1-1} E_{i_1}E_h\Big) = P\Big(\bigcup_{h=1}^{i_1-1} F_h\Big), \quad (F_h = E_{i_1}E_h)$$

$$(-\leq)^{j+1} \sum_{r=1}^{j}(-1)^{r+1} \sum_{h_1,h_2,h_r:\, h_r<h_2<h_1} P(F_{h_1}F_{h_2}F_{h_r})$$

$$= \sum_{r=1}^{j}(-1)^{r+1} \sum_{h_1,h_2,h_r:\, h_r<h_2<h_1<i_1} P(E_{i_1}E_{h_1}E_{h_2}E_{h_r})$$

$$= \sum_{r=1}^{j}(-1)^{r+1} \sum_{i_2,i_3,i_{r+1}:\, i_{r+1}<i_3<i_2<i_1} P(E_{i_1}E_{i_2}E_{i_3}E_{i_{r+1}})$$

$\therefore$ from 8.0,

$$P\Big(\bigcup_{i=1}^{n} E_i\Big) = \sum_{i_1=1}^{n} P(E_{i_1}) - \sum_{i_1=1}^{n} P\Big(\bigcup_{i_2:\, i_2<i_1} E_{i_1}E_{i_2}\Big)$$

$$(-\leq)^{j+2} \sum_{i_1=1}^{n} P(E_{i_1}) - \sum_{i_1=1}^{n}\sum_{r=1}^{j}(-1)^{r+1} \sum_{i_2,i_3,i_{r+1}:\, i_{r+1}<i_3<i_2<i_1} P(E_{i_1}E_{i_2}E_{i_{r+1}})$$

9.1 $P(\bigcup_{i=1}^{n} E_i)\ (-\leq)^{k+1} \sum_{r=1}^{k}(-1)^{r+1} \sum_{i_1,i_2,i_r:\, i_r<i_2<i_1} P(E_{i_1}E_{i_2}\cdots E_{i_r})$, $k\in Z^+$

here $(-\leq)^{k}$: $\begin{cases} \leq, & \text{if } k \text{ even} \\ \geq, & \text{if } k \text{ odd} \end{cases}$

$(-\geq)^{k}$: $\begin{cases} \geq, & \text{if } k \text{ even} \\ \leq, & \text{if } k \text{ odd} \end{cases}$

9.2 proof: (by induction)

let $k=1 \Rightarrow P(\bigcup E_i) \leq \sum^{n} P(E_i)$, $\forall n\in Z$.

let $k=2 \Rightarrow P(\bigcup E_i) \geq \sum_{i_1=1}^{n} P(E_{i_1}) - \sum_{i_1,i_2:\, i_2<i_1} P(E_{i_1}E_{i_2})$ $\forall n\in Z$

let $k=3 \Rightarrow P(\bigcup_{i=1}^{n} E_i) \leq \sum_{i_1=1}^{n} P(E_{i_1}) - \sum_{i_1,i_2:\, i_2<i_1} P(E_{i_1}E_{i_2}) + \sum_{i_1,i_2,i_3:\, i_3<i_2<i_1} P(E_{i_1}E_{i_2}E_{i_3})$, $\forall n\in Z$